Name:_____Period:_____

DISCOVER ESSRT 3

The Rock Cycle Infographic

Earth & Space Sciences Reference Tables, page 15 · HS-ESS2-5

Directions: Follow the instructions to work through each part. Answer the questions in the spaces provided. Parts marked **PAPER** use only your printed Reference Tables. Parts marked **EXPLORER** use the Rock Cycle Explorer on the class website.

Vocabulary: banding, cementation, compaction, deposition, erosion, extrusive, foliation, horizontal sorting, intrusive, lithification, magma, metamorphism, porphyry, sediment, weathering

Prior Knowledge Questions

Do these BEFORE you open your Reference Tables or the Explorer.

1.A sandcastle falls apart by the next morning, but sandstone cliffs can last millions of years. Both are made of sand. What do you think sandstone has that a sandcastle does not?

2.A chocolate bar left in a hot car gets soft and bends out of shape. Leave it long enough and it melts completely.

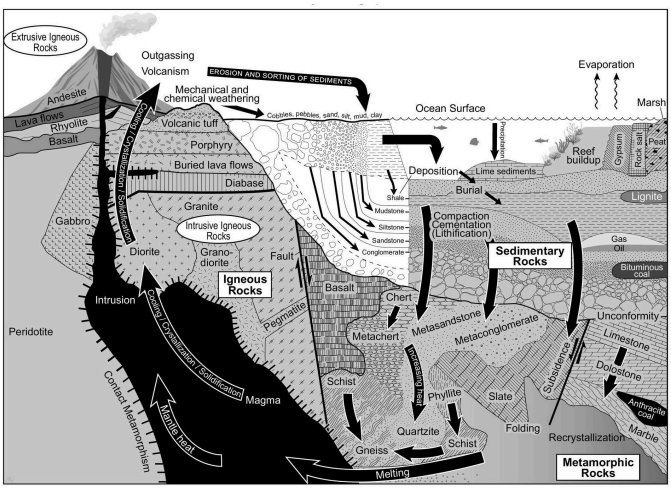
a.How might rock deep underground be similar to the chocolate?_____

b.How might it be different?_____

Warm-up

PAPER · No device needed

Open your printed Reference Tables to **page 15**, the Rock Cycle Infographic (shown below). This one page shows dozens of rocks and processes at once. Before using the Explorer, get to know the page.



ESSRT page 15: Rock Cycle Infographic (NYSED, 2026 Rev. Edition)

1.Three labels on the page are printed in bold inside boxes. These are the three **rock families**. Write them below.

1. _____ 2. _____ 3. _____

2.Find **Magma** near the bottom left. A large white arrow points up toward it from the bottom of the page. What does that arrow say?

Arrow label: _____

3.Look at the far right edge of the page outside the infographic. The arrow says “Increasing _____&_____”

b.Which way does it point? _____

Activity A — The Basic Rock Cycle

Get the Explorer ready:	Question:
Open the **Rock Cycle Explorer** (page 15) on the class website. Click **New to this? See the basic cycle first**. Use **Next** and **Back** to move through the steps.	What are the main parts of the rock cycle, and how does each one change into another?

EXPLORER

1.**Record:** Click through the walkthrough. Each step draws new arrows on the real page. For every arrow, record where it ends and the process written on it. The first row is done for you, and each arrow’s starting point is filled in.

#	From	To	Process on the arrow
1	Sediments	Sedimentary Rock	Compaction & Cementation
2	Sedimentary Rock		
3	Sedimentary Rock		
4	Metamorphic Rock		
5	Igneous Rock		
6	Igneous Rock		
7	Metamorphic Rock		
8	Magma		

2.**Analyze:** Look at the processes in your table.

a.Which process shows up on the most arrows? _____

b.How many times? ____

c.What does this tell you about which rocks can become sediment? _____

3.**Analyze:** In this walkthrough, which is the only rock family with an arrow pointing to **Magma**?

Rock family: _____

a.Why do you think rocks usually go through metamorphism before they melt? (Hint: think about what happens to a rock as it gets buried deeper and deeper.)

_____

Activity B — How Igneous Rock Forms

Get the Explorer ready:	Question:
(Click) **Close** any walkthrough that is open. Click **Zoom in: how igneous rock forms** (or click the **Igneous Rocks** label on the diagram).	How does the place where magma cools affect the rock it forms?

EXPLORER

1.**Observe:** Watch the magnifier closely in steps 4 and 5, then fill in the table.

Where it cools	Cooling speed	Crystal size	Example rock
Intrusive (step 4)			
Extrusive (step 5)			

2.**Explain:** Why does cooling slowly make bigger crystals? Use the word **atoms** in your answer.

(Circle the correct answers) *When magma cools slowly, atoms have* ***(less / more)*** *time to move around and arrange themselves into crystals, allowing* ***(smaller / larger)*** *crystals to grow.*

3.Read step 6. What is caught in the cooling lava to leave holes in the rock? _____. This creates a vesicular texture.

4.Read step 7. Porphyry has large crystals scattered in a fine-grained background. Why?

(Circle the correct answers) Sometimes magma starts cooling slowly deep underground, and a few crystals grow ***(small / large)***. Then the magma moves up closer to the surface, and the rest cools ***(slow / fast)***.

5.**Record:** Read step 8, “Same magma, two rocks.” Fill in the missing rock names.

Magma color	Cools SLOWLY (intrusive)	Cools FAST (extrusive)
Light-colored (felsic)		
Dark-colored (mafic)		

a.Across each row, what stays the same?_____

b.Down each column, what stays the same?_____

PAPER · Back to the page

6. Close the walkthrough and look at page 15. Why are **Basalt, Rhyolite, and Andesite** shown on the volcano, while **Granite, Diorite, and Gabbro** are shown underground?

_____

Activity C — How Sedimentary Rock Forms

Get the Explorer ready:	Question:
(Click) **Close** any walkthrough that is open. Click **Zoom in: how sedimentary rock forms**.	How does loose sediment become solid rock, and why do different sedimentary rocks form in different places?

EXPLORER

1.Read step 3. How do the pieces of rock change as they travel down the river?

(Circle the correct answers) The ***(biggest / smallest)*** pieces roll and bounce along the bottom, while the finest ones float along in the water. While they travel, the pieces bump into each other and get worn ***(sharper / rounder)*** and ***(larger / smaller)***.

2.**Predict:** Before you click to step 4: when the river reaches the sea and slows down, which pieces do you think will drop out first? Why?

_____

_____

3.**Observe:** Watch step 4. Write the sediment sizes in order from the shore to the deepest water.

Closest to shore			Farthest out

a. When the river reaches the sea, its water slows down and can no longer carry its load, so the sediment drops out. What is this process called? _____

4.Read steps 5 and 6. Matching: Match each process to its correct description.

_____ 1. Compaction _____ 2. Cementation _____ 3. Lithification	A. Dissolved minerals crystallize in the spaces between sediment grains, gluing them together. B. The weight of overlying sediment presses grains closer together, squeezing out water and making the layer thinner. C. The process of turning loose sediment into sedimentary rock through compaction and cementation.

5.**Record:** Use steps 6 and 7 to complete the table.

Sediment	Rock it becomes	What the rock looks or feels like
Sand		
Mud & clay		

PAPER · Back to the page

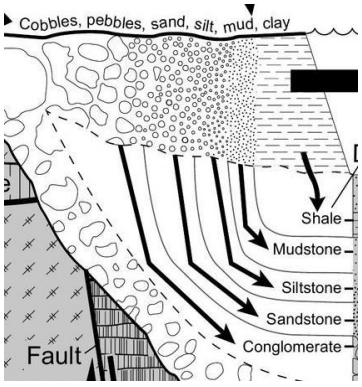
6.On your printed page, the sediment labels run **Cobbles, pebbles, sand, silt, mud, clay** from left to right. The rock names run from **Shale** at the top to **Conglomerate** at the bottom.

a.Which side of this part of the page is closest to the shore?

(circle) left or right

b.How do you know?_____

c.**<u>Draw one line on the picture</u>** connecting **pebbles** to the rock they become.



Detail of page 15: Erosion and sorting of sediments

EXPLORER

7. **Explore:** Not every sedimentary rock is made of broken pieces. (Click) **Close** the walkthrough and click each rock below. Record how it formed.

Rock	How it formed
Limestone	
Gypsum	

a. Gypsum's card has no **Trace this rock's path** button. Why? (read the description)

_____

Activity D — How Metamorphic Rock Forms

Get the Explorer ready:	Question:
(Click) **Close** any walkthrough that is open. Click **Zoom in: how metamorphic rock forms**. Watch the heat and pressure gauges as you go.	How do heat and pressure change a rock without melting it?

EXPLORER

1. Read step 1. What do the word parts mean? "Meta" means: _____ "Morph" means: _____

2. **Record:** Move through steps 2–6. For each rock, describe what is happening to its mineral grains and how you could recognize it.

Rock (heat & pressure increase ↓)	What happens to the grains	How to recognize it
Slate		
Phyllite		
Schist		
Gneiss		

3. Read step 3. What is **foliation**?_____

a. Why does slate split into flat sheets?_____

4. Read step 7.

a. What happens if the rock gets even hotter?_____

b It isn't metamorphic anymore. When magma cools, it becomes an _____ rock.

5. **Record:** Read step 8, "Other rocks change too." Fill in the metamorphic rock each one becomes.

Parent rock	Metamorphic rock
Sandstone	
Limestone or dolostone	
Bituminous coal	

a.Most of these rocks do not form bands like gneiss. Why not?_____

6.**Try it:** (Click) **Close** the walkthrough. Click **Gneiss** and press **Trace this rock's path**.

a.Write each rock or material in the path, in order.

_____

b.Where does the path end? _____

c.Why is this the best evidence on the page that rocks move in a cycle and not a straight line?_____

_____

PAPER · Back to the page

7.The word **Schist** is printed in two places on page 15. Find both on your printed page.

a.What rock does each Schist form from? (Look at what is drawn right above each one, and follow the black arrows.)

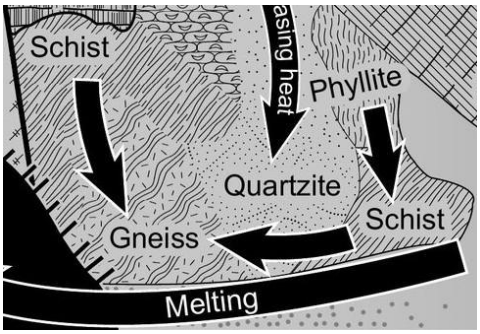
Schist #1 forms from: _____

Schist #2 forms from: _____

b.Does a metamorphic rock always come from the same rock family? Explain.

_____

_____



Detail of page 15: the lower middle of the page

Build It Better

Your Feedback on the Explorer

The Explorer is a **model** of your printed Reference Tables: the same data, shown a new way. Scientists and engineers test their models, find the strengths and limitations, and improve them. That's your job on this page. Your teacher is working with a company that builds the Explorer by coding, and feedback like yours is what tells it what to change in the next version.

No name needed.

Part of the Explorer: Rock Cycle Explorer (page 15)

Device: □ Chromebook □ Laptop □ Phone □ Tablet

1. **Rate it.** Circle one number for each.

	1 No	2	3	4 Yes!
It was easy to figure out what to click.	1	2	3	4
It helped me understand the printed page.	1	2	3	4
It worked well on the device I used.	1	2	3	4
I would use it again to study.	1	2	3	4

2. **Two models, one set of data.** Compare the Explorer with the printed page.

+ **Strength.** Something the Explorer shows **better** than the printed page:	– **Limitation.** Something the printed page does better, or something the Explorer **gets wrong or leaves out**:

3. **Fix one thing.** Be specific enough that someone could build it from what you write.

What kind?	□ Something broke □ Something was confusing □ An idea for something new
Where? Which step, screen, or button	
What happened? Or: what's missing	
What should happen instead?	

4. **Design it.** Sketch your change as it would look on the screen, and label it.

For teachers: NYSSLS Science & Engineering Practices: Developing and Using Models (evaluate the merits and limitations of a model) and Engineering Design (evaluate a solution against criteria; refine it). HS-ETS1-3.